

ANALYSIS I: LIMITS AND CONTINUITY

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- This problem set deals with *limits* and *continuity*.
- Numericals: These problems are a chance for you to familiarise yourself with computations and key examples and non-examples.
- Prove Or Disprove And Salvage If Possible: Some of the statements here are true, while others are not. If a statement is true, prove it and see if it can be strengthened. If it is false, give a counterexample and prove a salvage of the statement.
- Exploration: An opportunity to prove a larger mathematical result.

NUMERICALS

Problem 1. Determine, with proof, if the following limits exist. If they exist, compute them.

$$(a) \lim_{x \rightarrow 1} \frac{1}{x} \quad (b) \lim_{x \rightarrow 0} \frac{\lfloor x \rfloor}{x} \quad (c) \lim_{x \rightarrow \infty} \sin(x)$$

Problem 2. Prove the following:

- (a) If f and g are continuous at $x \in \mathbb{R}$, then so is $f + g$.
- (b) If f is continuous at $x \in \mathbb{R}$, then so is cf for any $c \in \mathbb{R}$.
- (c) For any $n \in \mathbb{N}$, the function $f(x) = x^n$ is everywhere continuous.
- (d) Prove that the function

$$f(x) = 10x^9 - 87x^6 + 543x^2 + 1$$

is everywhere continuous. Can you generalise?

- (e) Prove that if f is continuous at $x \in \mathbb{R}$ and g is continuous at $f(x)$, then $g \circ f$ is continuous at x .

Problem 3. Give examples of the following:

- (a) A function that is nowhere continuous.
- (b) A continuous function $g : \mathbb{R} \rightarrow \mathbb{R}$ such that every open interval $(-\varepsilon, \varepsilon)$ with $\varepsilon > 0$ contains infinitely many points where g is zero and infinitely many points where g is nonzero.
- (c) A function $f : \mathbb{R} \rightarrow \mathbb{R}$ is said to have the *intermediate value property* if for any $x, y, c \in \mathbb{R}$ with $f(x) < c < f(y)$, there exists $z \in \mathbb{R}$ between x and y such that $f(z) = c$. Give an example of a discontinuous function with the intermediate value property. Can you find a non-decreasing example?

PROVE OR DISPROVE AND SALVAGE IF POSSIBLE (PODASIPS)

Problem 4. Prove that a function $f : \mathbb{R} \rightarrow \mathbb{R}$ is continuous at $x \in \mathbb{R}$ if and only if $f(x_n) \rightarrow f(x)$ for every sequence $x_n \rightarrow x$.

Problem 5. Prove that if $f, g : \mathbb{R} \rightarrow \mathbb{R}$ are functions such that

$$\lim_{x \rightarrow a} f(x) = b \quad \text{and} \quad \lim_{y \rightarrow b} g(y) = c$$

then

$$\lim_{x \rightarrow a} g(f(x)) = c.$$

Be careful!

Problem 6. A function $f : \mathbb{R} \rightarrow \mathbb{R}$ is continuous if and only if

$$\lim_{h \rightarrow 0} (f(x+h) - f(x-h)) = 0$$

holds for all $x \in \mathbb{R}$.

Problem 7. In this problem, you may use that any open interval in $\mathbb{R}$ contains a rational number.

- (a) Prove that if $f : \mathbb{R} \rightarrow \mathbb{R}$ is a continuous function such that $f(x) = 0$ for all $x \in \mathbb{Q}$ then $f(x) = 0$ for all $x \in \mathbb{R}$.
- (b) Prove that the function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$f(x) = \begin{cases} 0 & \text{if } x \in \mathbb{Q}, \\ x & \text{if } x \notin \mathbb{Q} \end{cases}$$

is everywhere discontinuous.

Problem 8. Suppose that $f : \mathbb{R} \rightarrow \mathbb{R}$ is monotonic and

$$\lim_{x \rightarrow \infty} \frac{f(2x)}{f(x)} = 1.$$

Prove that for any $c > 0$, we have that

$$\lim_{x \rightarrow \infty} \frac{f(cx)}{f(x)} = 1.$$

Problem 9. Suppose that $f, g : \mathbb{R} \rightarrow \mathbb{R}$ are continuous functions such that

$$\lim_{x \rightarrow \infty} (f(x) - g(x)) = 0.$$

Does it follow that $f = g$? What if f, g are periodic with rational periods? What if f, g are periodic with possibly irrational periods?

EXPLORATION

Problem 10. In this problem, we will explore so-called contractions. Given a subset $D \subseteq \mathbb{R}$, we say that a function $f : D \rightarrow D$ is a *contraction mapping* if there exists $0 \leq \lambda < 1$ such that for all $x, y \in D$:

$$|f(x) - f(y)| \leq \lambda \cdot |x - y|.$$

Any λ with this property shall henceforth be referred to as a *Lipschitz constant* of f .

We say that $x \in D$ is a *fixed point* of a function $f : D \rightarrow D$ if $f(x) = x$. This problem is about understanding the fixed points of a contraction mapping.

- (a) Determine for which $a \in \mathbb{R}$ the function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = ax$ is a contraction mapping.
- (b) For which $0 < \varepsilon < 1$ is the function $f : [-\varepsilon, \varepsilon] \rightarrow [-\varepsilon, \varepsilon]$ defined by $f(x) = x^2$ a contraction mapping?
- (c) Does there exist a function $f : D \rightarrow D$ such that for all $x, y \in D$ with $x \neq y$ we have that

$$|f(x) - f(y)| < |x - y|$$

yet f is *not* a contraction mapping? We shall call any such function *weakly contracting*.

- (d) Prove that any contraction mapping is continuous.

- (e) Prove that a contraction mapping can have at most one fixed point.
 (f) Let $a \in D$ where $D \subseteq \mathbb{R}$ is a closed set and let $f : D \rightarrow D$ be a contraction mapping with Lipschitz constant λ . Consider the recursive sequence defined by

$$\begin{aligned} a_0 &= a \\ a_n &= f(a_{n-1}) \end{aligned} \quad \text{for } n \geq 1.$$

Using that f is a contraction mapping prove that

$$|a_{n+1} - a_n| \leq \lambda |a_n - a_{n-1}|$$

for any $n \geq 1$. Deduce that

$$|a_{n+m} - a_n| \leq \frac{\lambda^n}{1 - \lambda} \cdot |a_1 - a_0|$$

for all $m, n \geq 1$.

- (g) A sequence of real numbers b_n is said to be *Cauchy* if for every $\varepsilon > 0$ there exists $N > 0$ such that for all $n, m \geq N$:

$$|b_n - b_m| < \varepsilon.$$

It is a fact of the real numbers that any Cauchy sequence is convergent. Use this together with the fact that D is closed to conclude that a_n has a limit point $c \in D$.

- (h) Deduce from part (d) and part (g) that c is a fixed point of f .

Putting everything together, you have just proved the following fixed-point theorem:

Theorem. *Let $D \subseteq \mathbb{R}$ be a closed set. If $f : D \rightarrow D$ is a contraction mapping, then f has exactly one fixed point.*

Let us finish by considering whether all the hypotheses of this theorem are necessary.

- (i) Does the theorem still hold if D is an open interval?
 (j) Assume instead that f is weakly contracting as defined in part (c). Does the uniqueness part of the theorem still hold? Does the existence part of the theorem still hold? What if D is bounded?